

# Cavitoon Lite Quick Start

Install and verify World cavity in a Unity 6 URP project

## 1. Requirements

- Unity 6 with the Universal Render Pipeline.
- URP Render Graph enabled. Compatibility Mode is not supported.
- At least one opaque object using a standard URP material for the first test.

## 2. Automatic Installation

1. In the Project window, select the **Universal Renderer Data** used by your active URP Pipeline Asset. It is commonly named UniversalRendererData or ForwardRenderer.
2. Run **Tools > Cavitoon > Install Into Selected URP Renderer**.
3. Select the added **Cavitoon Cavity Renderer Feature** in the Renderer Data Inspector.
4. Keep **Injection Point** set to **Before Rendering Transparents**.
5. Enter Play mode and inspect opaque geometry in the Game view.

The installer does not add a duplicate if the selected Renderer Data already contains Cavitoon. Keep Show In Scene View enabled if you also want an Editor preview.

## 3. Recommended First Settings

| Setting                | Value                         |
|------------------------|-------------------------------|
| World Radius           | 0.25                          |
| World Ridge            | 1.0                           |
| World Valley           | 1.0                           |
| World Attenuation      | 1.0                           |
| World Sample Count     | 8                             |
| Max World Pixel Radius | 24                            |
| Injection Point        | Before Rendering Transparents |

World Radius is measured in scene units. If the effect is barely visible, first check the scale of your objects, then adjust Radius, Ridge, and Valley.

## 4. Manual Installation

1. Open the active **Universal Renderer Data** asset in the Inspector.
2. Under **Renderer Features**, click **Add Renderer Feature**.
3. Select **Cavitoon Cavity Renderer Feature**.
4. Confirm that the camera uses the Pipeline Asset and Renderer Data where Cavitoon was installed.
5. Leave the Shader field assigned to Hidden/Cavitoon/Cavity. No material needs to be created manually.
6. Enter Play mode and verify the result on opaque geometry.

## 5. Verify the Setup

- Ridge should brighten convex details and visible outer contours.
- Valley should darken recessed areas, intersections, and joints.
- Setting both Ridge and Valley to 0 should make the frame unchanged.
- Changing World Sample Count should affect quality and GPU cost, not the search radius.
- The original camera alpha should remain unchanged.

## If the Effect Is Not Visible

- Confirm that the active camera uses the Renderer Data containing Cavitoon.
- Make sure the Renderer Feature is enabled.
- Disable URP Render Graph Compatibility Mode.
- Test with an opaque standard URP Lit material.
- Increase World Ridge or World Valley temporarily.
- Match World Radius to the object scale.
- Confirm that the Shader field references Hidden/Cavitoon/Cavity.
- Check the Console for missing color, depth, or normals warnings.

Cavitoon requests camera depth and normals automatically. URP may add a depth/normals prepass depending on the renderer and platform. The global Opaque Texture setting is not required.

For parameter descriptions, limitations, performance guidance, and advanced troubleshooting, open **ReadMe.pdf**.